J.N.T.U.H UNIVERSITY COLLEGE OF ENGINEERING HYDERABAD

(Autonomous)

KUKATPALLY, HYDERABAD - 500 085



Certificate

Certified that this is the bonafide record of the practical work done during the academic year _____ by

Name _____

Roll Number _____ Class _____

In the Laboratory of _____

of the Department of _____

Signature of the Staff Member ***Signature of the Head of the Department***

Date of Examination_____

Signature of the Examiner/s

Internal Examiner ***External Examiner*** q

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List of Experiments

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2	Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should AND, OR and XOR each character in this string with 127 and display the result.				
3	Write C/Java program to perform encryption and decryption using Ceaser cipher				
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1. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should XOR each character in this string with 0 and display the result.

```
#include <stdio.h>

int main()

{

   char *str = "Hello World";

   printf("Original String:%s\n", str);

   printf("After XOR with 0:");

   for (int i = 0; str[i] != '\0'; i++)

   {

      char ch = str[i] ^ 0;

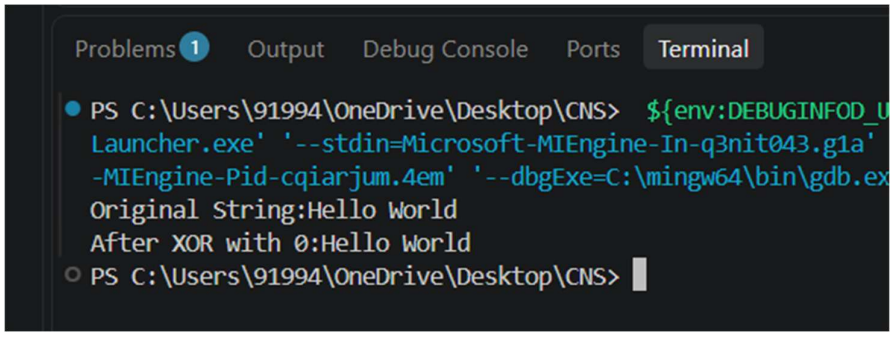
      printf("%c", ch);

   }

   return 0;

}
```

OUTPUT:



2. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should AND, OR and XOR each character in this string with 127 and display the result.

```
#include <stdio.h>

int main()

{

   char *str = "Hello World";

   printf("Original String:%s\n", str);
```

```
  printf("After AND with 127:");
  for (int i = 0; str[i] != '\0'; i++)
  {
    char ch = str[i] & 127;
    printf("%c", ch);
  }
  printf("\nAfter XOR with 127:");
  for (int i = 0; str[i] != '\0'; i++)
  {
    char ch = str[i] ^ 127;
    printf("%c", ch);
  }
  printf("\nAfter OR with 127:");
  for (int i = 0; str[i] != '\0'; i++)
  {
    char ch = str[i] | 127;
    printf("%c", ch);
  }
  return 0;
}
```

OUTPUT:

```
Original String:Hello World
After AND with 127:Hello World
After XOR with 127:■■■■ø■■■ø■ā
After OR with 127:ÿÿÿÿÿ ÿÿÿÿÿ
```

3. Write C/Java program to perform encryption and decryption using Ceaser cipher.

```
import java.util.Scanner;
public class Caesar{
  public static String encrypt(String text,int key){
```

```
    String result="";
    key=((key%26)+26)%26;
    for(int i=0;i<text.length();i++){
      char c=text.charAt(i);
      if(c>='A' && c<='Z'){
        c=(char)('A'+(c-'A'+key)%26);
      }else if(c>='a'&& c<='z'){
        c=(char)('a'+(c-'a'+key)%26);
      }
      result=result+c;
    }
    return result;
  }
  public static String decrypt(String text,int key){
    return encrypt(text,-key);
  }
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    System.out.print("enter plain text ");
    String text=sc.nextLine();
    System.out.print("enter key");
    int key=sc.nextInt();
    String encrypted=encrypt(text,key);
    System.out.print("encrypted text is "+ encrypt(text,key));
    System.out.print("decrypted text is "+ decrypt(encrypted,key));
  }
}
```

OUTPUT:

```
enter plain text Hello
enetr key3
encrypted text is Khoordecrypted text is Hello
```

4. Write C/Java program to perform encryption and decryption using play fair cipher

```
import java.util.Scanner;
public class Playfair{
  static char[][] mat=new char[5][5];
  static int[] row=new int[26];
  static int[] col=new int[26];
  public static void creatematrix(String key){
    boolean[] used=new boolean[26];
    used['J'-'A']=true;
    String text=key.toUpperCase().replace('J','I')+"ABCDEFGHIKLMNOPQRSTUVWXYZ";
    int index=0;
    for(int i=0;i<text.length();i++){
      char c=text.charAt(i);
      if(c<'A'||c>'Z'){continue;}
      int letter=c-'A';
      if(!used[letter]){
        used[letter]=true;
        mat[index/5][index%5]=c;
        row[letter]=index/5;
        col[letter]=index%5;
        index++;
      }
    }
  }
  public static String preparepair(String text){
    text=text.toUpperCase().replace('J','I');
    String result="";
    int i=0;
    while(i<text.length()){
```

```
        char first=text.charAt(i);
        result+=first;
        i++;
        if(i==text.length()){
            result+='X';
        }else if(first==text.charAt(i)){
            result+='X';
        }else{
            result+=text.charAt(i);
            i++;
        }
    }
    return result;
}
public static String process(String text,boolean encrypt){
    String result="";
    int direction=encrypt?1:4;
    for(int i=0;i<text.length();i+=2){
        char a=text.charAt(i);
        char b=text.charAt(i+1);
        int r1,r2,c1,c2;
        r1=row[a-'A'];
        r2=row[b-'A'];
        c1=col[a-'A'];
        c2=col[b-'A'];
        if(r1==r2){
            c1=(c1+direction)%5;
            c2=(c2+direction)%5;
        }else if(c1==c2){
            r1=(r1+direction)%5;
```

```
            r2=(r2+direction)%5;
        }else{
            int temp=c1;
            c1=c2;
            c2=temp;
        }
        result+=mat[r1][c1];
        result+=mat[r2][c2];
    }
    return result;
}
public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the keyword: ");
    String key = sc.nextLine();
    creatematrix(key);
    System.out.print("Enter the plaintext: ");
    String plaintext = sc.nextLine();
    String prepared = preparepair(plaintext);
    String encrypted = process(prepared, true);
    String decrypted = process(encrypted, false);
    System.out.println("\nKey Matrix:");
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
            System.out.print(mat[i][j] + " ");
        }
        System.out.println();
    }
    System.out.println("\nPrepared text: " + prepared);
    System.out.println("Encrypted text: " + encrypted);
```

```
        System.out.println("Decrypted text: " + decrypted);
        sc.close();
    }
}
```

OUTPUT:

```
Enter keyword: MONARCHY
Enter plaintext: INSTRUMENT
Prepared text: INSTRUMENT
Encrypted text: GATLMZCLRQ
Decrypted text: INSTRUMENT
```

5. Write C/Java program to perform encryption and decryption using Hill Cipher.

```
import java.util.Scanner;
public class Hill{
    static int[][] key={
        {3,3},
        {2,5}
    };
    static int[][] inversekey={
        {15,17},
        {20,9}
    };
    public static String process(String text,int[][] mat){
        StringBuilder result=new StringBuilder();
        for(int i=0;i<text.length();i+=2){
            int a=text.charAt(i)-'A';
            int b=text.charAt(i+1)-'A';
            int c1=(mat[0][0]*a+mat[0][1]*b)%26;
            int c2=(mat[1][0]*a+mat[1][1]*b)%26;
            result.append((char)('A'+c1));
            result.append((char)('A'+c2));
        }
```

```
        return result.toString();
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the plaintext: ");
        String plaintext = sc.nextLine();
        plaintext = plaintext.toUpperCase();
        if (plaintext.length() % 2 != 0) {
            plaintext += "X";
        }
        String encrypted = process(plaintext, key);
        String decrypted = process(encrypted, inversekey);
        System.out.println("Prepared plaintext: " + plaintext);
        System.out.println("Encrypted text: " + encrypted);
        System.out.println("Decrypted text: " + decrypted);
        sc.close();
    }
}
```

OUTPUT:

```
Enter the plaintext: MEET
Prepared plaintext: MEET
Encrypted text: RHHX
Decrypted text: MEET
```

6. Write a C/JAVA program to implement the DES algorithm logic.

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import java.util.Base64;
import java.util.Scanner;
public class DESExample {
    public static void main(String[] args) throws Exception {
```

```
    Scanner sc = new Scanner(System.in);
    KeyGenerator keyGen =KeyGenerator.getInstance("DES");
    SecretKey secretKey = keyGen.generateKey();
    Cipher cipher =Cipher.getInstance("DES/ECB/PKCS5Padding");
    System.out.print("Enter Plain Text: ");
    String message = sc.nextLine();
    cipher.init(Cipher.ENCRYPT_MODE, secretKey);
    byte[] encrypted = cipher.doFinal(message.getBytes());
    String encryptedText = Base64.getEncoder().encodeToString(encrypted);
    System.out.println("Original Message: " + message);
    System.out.println("Encrypted Message: " + encryptedText);
    cipher.init(Cipher.DECRYPT_MODE, secretKey);
    byte[] decrypted = cipher.doFinal(Base64.getDecoder().decode(encryptedText));
    String decryptedText = new String(decrypted);
    System.out.println("Decrypted Message: " + decryptedText);
    sc.close();
  }
}
```

OUTPUT:

Original Text: HELLO WORLD

Encrypted Text: bRynoU3KYlRZY75h2h2xkA==

Decrypted Text: HELLO WORLD

7. Write a C/JAVA program to implement the Blowfish algorithm logic

```
public import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import java.util.Base64;
import java.util.Scanner;
```

```
public class BlowfishExample {
   public static void main(String[] args) throws Exception {
      Scanner sc = new Scanner(System.in);
      KeyGenerator keyGen = KeyGenerator.getInstance("Blowfish");
      keyGen.init(128);
      SecretKey secretKey = keyGen.generateKey();
      Cipher cipher = Cipher.getInstance("Blowfish");
      System.out.print("Enter Plain Text: ");
      String message = sc.nextLine();
      cipher.init(Cipher.ENCRYPT_MODE, secretKey);
      byte[] encrypted = cipher.doFinal(message.getBytes());
      String encryptedText = Base64.getEncoder().encodeToString(encrypted);
      System.out.println("Original Message: " + message);
      System.out.println("Encrypted Message: " + encryptedText);
      cipher.init(Cipher.DECRYPT_MODE, secretKey);
      byte[] decrypted = cipher.doFinal(Base64.getDecoder().decode(encryptedText));
      String decryptedText = new String(decrypted);
      System.out.println("Decrypted Message: " + decryptedText);
      sc.close();
   }
}
```

OUTPUT:

```
Original Text: HELLO WORLD
Encrypted Text: C19JFJy/nkC7hNZNAQMiuw==
Decrypted Text: HELLO WORLD
```

8. Write a C/JAVA program to implement the Rijndael algorithm logic

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
```

```
import javax.crypto.SecretKey;
import java.util.Base64;
import java.util.Scanner;
public class RijndaelExample {
   public static void main(String[] args) throws Exception {
      Scanner sc = new Scanner(System.in);
      KeyGenerator keyGen = KeyGenerator.getInstance("AES");
      keyGen.init(128);
      SecretKey secretKey = keyGen.generateKey();
      Cipher cipher = Cipher.getInstance("AES");
      System.out.print("Enter Plain Text: ");
      String message = sc.nextLine();
      cipher.init(Cipher.ENCRYPT_MODE, secretKey);
      byte[] encrypted = cipher.doFinal(message.getBytes());
      String encryptedText = Base64.getEncoder().encodeToString(encrypted);
      System.out.println("Original Message: " + message);
      System.out.println("Encrypted Message: " + encryptedText);
      cipher.init(Cipher.DECRYPT_MODE, secretKey);
      byte[] decrypted = cipher.doFinal(Base64.getDecoder().decode(encryptedText));
      String decryptedText = new String(decrypted);
      System.out.println("Decrypted Message: " + decryptedText);
      sc.close();
   }
}
```

OUTPUT:

```
Original Text: HELLO WORLD
Encrypted Text: ewJieTvTN7ehW6Vm3wVVdg==
Decrypted Text: HELLO WORLD
```

9. Write C/Java program to implement the RSA algorithm

```
import java.util.Scanner;
public class rsa {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter p: ");
      int p = sc.nextInt();
      System.out.print("Enter q: ");
      int q = sc.nextInt();
      int n = p * q;
      int phi = (p - 1) * (q - 1);
      System.out.print("Enter e: ");
      int e = sc.nextInt();
      int d = 1;
      while ((d * e) % phi != 1) {
         d++;
      }
      System.out.print("Enter message: ");
      int m = sc.nextInt();
      int c = 1;
      for (int i = 0; i < e; i++) {
         c = (c * m) % n;
      }
      int decrypted = 1;
      for (int i = 0; i < d; i++) {
         decrypted = (decrypted * c) % n;
      }
      System.out.println("n = " + n);
      System.out.println("phi(n) = " + phi);
      System.out.println("Public Key: (" + e + ", " + n + ")");
      System.out.println("Private Key: (" + d + ", " + n + ")");
```

```
        System.out.println("Encrypted Message: " + c);
        System.out.println("Decrypted Message: " + decrypted);
        sc.close();
    }
}
```

OUTPUT:

```
Enter p: 61
Enter q: 53
Enter e: 17
Enter message: 65
n = 3233
phi(n) = 3120
Public Key: (17, 3233)
Private Key: (2753, 3233)
Encrypted Message: 2790
Decrypted Message: 65
```

10. Implement the Diffie-Hellman Key Exchange mechanism in Java

```
import java.util.Scanner;
public class Diffiehellmankeyexchange{
   public static void main(String[] args){
      Scanner sc=new Scanner(System.in);
      int p,g,a,b,Pa,Pb,Ka,Kb;
      p=23;
      g=5;
      System.out.print("enter private key of a:");
      a=sc.nextInt();
      System.out.print("enter private key of b:");
      b=sc.nextInt();
      Pa=(int)(Math.pow(g,a)%p);
      Pb=(int)(Math.pow(g,b)%p);
      Ka=(int)(Math.pow(Pb,a)%p);
      Kb=(int)(Math.pow(Pa,b)%p);
      System.out.println("Public prime "+p);
      System.out.println("Public generator "+g);
```

```
    System.out.println("Alice public key is "+Pa);
    System.out.println("Bobs public key is "+Pb);
    System.out.println("Shared key of Alice "+Ka);
    System.out.println("Shared key of Bob "+Kb);
    if(Ka==Kb){
      System.out.println("Key exchange successful!");
    }else{
      System.out.println("Key exchange failure");
    }
    sc.close();
```

OUTPUT:

```
enter private key of a:6
enter private key of b:15
Public prime 23
Public generator 5
Alice public key is 8
Bobs public key is 19
Shared key of Alice 2
Shared key of Bob 2
Key exchange successful!
```

11. Calculate the message digest of a text using the SHA-1 algorithm in Java

```
import java.security.MessageDigest;
import java.util.Scanner;
public class SHA1Example {
   public static void main(String[] args) throws Exception {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter the text: ");
      String message = sc.nextLine();
      MessageDigest md = MessageDigest.getInstance("SHA-1");
      byte[] digest = md.digest(message.getBytes());
      StringBuilder hexString = new StringBuilder();
      for (byte b : digest) {
         hexString.append(String.format("%02x", b & 0xff));
```

```
    }
    System.out.println("Original Message: " + message);
    System.out.println("SHA-1 Message Digest: " + hexString);
    sc.close();
  }
}
```

OUTPUT:

Enter the text: Hello

Original Message: Hello

SHA-1 Message Digest: f7ff9e8b7bb2e09b70935a5d785e0cc5d9d0abf0

12. Calculate the message digest of a text using the MD5 algorithm in JAVA

```
import java.security.MessageDigest;
import java.util.Scanner;
public class MD5Example {
  public static void main(String[] args) throws Exception {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the text: ");
    String message = sc.nextLine();
    MessageDigest md = MessageDigest.getInstance("MD5");
    byte[] digest = md.digest(message.getBytes());
    StringBuilder hexString = new StringBuilder();
    for (byte b : digest) {
      hexString.append(String.format("%02x", b & 0xff));
    }
    System.out.println("Original Message: " + message);
    System.out.println("MD5 Message Digest: " + hexString);
    sc.close();
```

```
    }
}
```

OUTPUT:

```
Enter the text: Hello

Original Message: Hello

MD5 Message Digest: 8b1a9953c4611296a827abf8c47804d7
```